

Stories We Tell (and the Machines Retell)

Cross-Cultural Narrative Structure in Llama 3.1 8B

NLP Final Project P12

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Abstract

This project investigates whether a large language model preserves the *structure* of a folktale when asked to retell it “in the style of” a different cultural tradition. Starting from a corpus of 30 folktales drawn equally from three traditions: European, Asian, and African Diaspora. We use Llama 3.1 8B (served locally via Ollama) to produce 60 cultural retellings. Each story version is converted into a *narrative graph* in which characters and objects are nodes and plot events are directed, labelled edges. We compare originals against retellings along four complementary dimensions: graph edit distance, relational motif overlap, character-role alignment, and sentence-embedding cosine similarity. Results show that (i) the original stories do not cluster cleanly by culture, suggesting shared structural archetypes; (ii) European-style retellings diverge most from the originals structurally; and (iii) African Diaspora-style retellings are closest to the originals on every metric, suggesting the model imposes weaker stylistic changes when targeting that tradition.

1 Introduction

Folktales are one of the most studied objects in computational narratology [1, 2]: they are short, self-contained, and exhibit recurring structural patterns that cross cultural boundaries. At the same time, each cultural tradition inflects those patterns in characteristic ways through the choice of characters, the type of conflict, and the nature of the resolution.

Large language models such as Llama 3.1 8B have absorbed vast multilingual text, including folktale collections, mythologies, and literary adaptations. When such a model is instructed to retell a story “in the style of” a particular cultural tradition, two outcomes are possible. Either the model reproduces the deep narrative structure faithfully while only adjusting surface elements (names, settings, vocabulary), or it restructures the plot itself in culturally typical ways. Distinguishing between these two outcomes requires a structural, not merely lexical, comparison.

This project operationalises that comparison through narrative graphs and four structural metrics, applied uniformly across 90 story versions (30 originals and 60 retellings). The pipeline combines prompt engineering with structured JSON output, local LLM inference via Ollama, and transformer-based sentence embeddings for semantic similarity.

2 Dataset

The corpus consists of 30 English-language folktales, 10 per cultural tradition: **European**, **Asian**, and **African Diaspora**. All texts are drawn from public-domain collections published

before 1930 (e.g. Fansler’s *Filipino Popular Tales*, Bompas’s *Folklore of the Santal Parganas*, and Carey’s *Fairy Legends of the French Provinces*). Story lengths range from roughly 320 to 900 words; the median is approximately 560 words.

Each of the 30 originals is retold into the two *other* cultural styles, yielding 60 retellings and a total of 90 story versions. Because the entire pipeline runs in English on a single local model, cross-lingual effects are deliberately excluded.

3 Methodology

3.1 Retelling with Llama 3.1 8B

Retellings are generated by `src/retell_all.py` using zero-shot prompting:

*Retell the following folktale in the style of {target} folktales.
Keep the SAME main events and the SAME ending.
Do NOT add new major plot points.
Output ONLY the retold story text. No title. No explanations.*

Temperature is set to 0.35 to allow stylistic variation while limiting hallucination; if the output is shorter than 200 words, up to two retries are made at higher temperatures (0.55, then 0.6). Each retelling targets a word count within $\pm 15\%$ of the original. For the African Diaspora target, the prompt additionally requires Standard English and forbids dialect or phonetic-spelling imitation, so that cultural style is expressed through themes, imagery, and oral-storytelling cadence rather than caricature. Ollama is used as the inference backend, communicating with the model via its local REST API.

3.2 Event Extraction

`src/extract_events_all.py` prompts the same model (temperature 0.2) to produce a JSON list of 10–14 chronological plot events per story version. The prompt enforces a strict schema:

```
{ "events": [ {"i": 1, "event": "..."}, ... ] }
```

A two-stage fallback handles malformed outputs: a retry with a stricter reminder prompt, followed by a regex parser that recovers event strings directly from partially-formed JSON. The output file `events_final.jsonl` contains 90 records covering all story versions.

3.3 Narrative Graph Construction

`src/extract_triples.py` converts each event list into (subject, relation, object) triples, again via structured LLM prompting. The relation phrase is constrained to 2–4 lowercase words, entity labels are normalised (articles stripped, lowercased), and the same character must receive the same label throughout a story.

`src/build_graphs.py` assembles these triples into directed multigraphs (NetworkX `MultiDiGraph`). Events with no clear object become self-loops. Within each story version, the entity appearing as subject most frequently is labelled `PROTAGONIST`; the runner-up is `DEUTERAGONIST`; everything else is `OTHER`.

From each graph we extract eight structural features: node count, edge count, self-loop count, distinct relation count, average out-degree, maximum out-degree, graph density, and number of weakly connected components.

3.4 Metrics

Four metrics compare an original story to each of its retellings:

Normalised graph edit distance (GED). We compute `networkx.graph_edit_distance` on the unlabelled digraphs (collapsing parallel edges) and normalise by $\max(|V| + |E|)$ so scores are comparable across stories of different lengths. A timeout of 10 seconds is imposed; unresolved pairs are excluded.

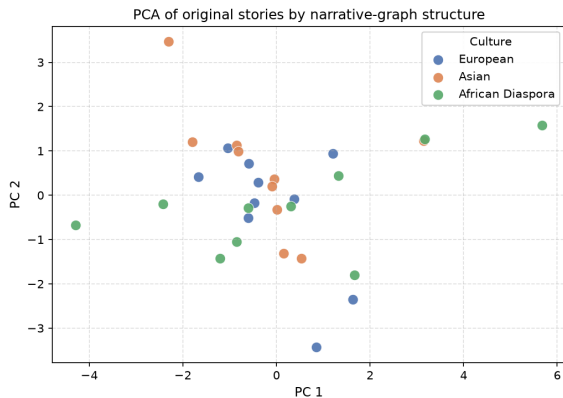
Relation/motif overlap. Jaccard similarity between the sets of action words in the original’s and retelling’s relation phrases (stop-words removed).

Role alignment. For each event position up to $\min(|S_1|, |S_2|)$, we compare whether the subject (and separately, the object) carries the same role label (PROTAGONIST, DEUTERAGONIST, or OTHER) in both versions. The score is the fraction of matching positions.

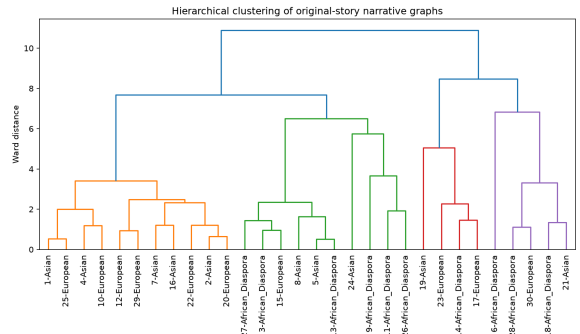
Semantic similarity. We encode each full story text with `all-MiniLM-L6-v2` [3], a BERT-based sentence encoder, and report the cosine similarity between the original and the retelling embeddings. This metric captures meaning-level preservation independently of graph structure.

4 Results

4.1 Clustering of Original Stories



(a) PCA of the 8 structural graph features (originals only).



(b) Hierarchical clustering dendrogram (Ward linkage).

Figure 1: Structure of the original-story space. In (a) colours indicate cultural tradition; in (b) leaf labels give story id and culture. Neither K-Means ($k=3$) nor hierarchical clustering recovers clean culture-level groups.

Figure 1 shows the structural feature space of the 30 original stories. K-Means ($k=3$) assigns 14, 13, and 3 stories to the three clusters, none of which aligns with a single culture (Table 1a). The dominant cluster (14 stories) contains 6 Asian, 6 European, and 2 African Diaspora texts. Hierarchical clustering (Ward linkage) produces a similarly mixed dominant group of 21 stories (Table 1b). These results provide mild evidence for universal narrative archetypes: at the level of simple graph statistics, folktales from different traditions occupy the same structural space.

Cluster	Eur.	Asian	A.D.
0	6	6	2
1	4	3	6
2	0	1	2

(a) K-Means ($k=3$).

Cluster	Eur.	Asian	A.D.
1	7	8	6
2	2	1	1
3	1	1	3

(b) Hierarchical / Ward ($k=3$).

Table 1: Culture breakdown per cluster for the 30 original stories. A.D. = African Diaspora.

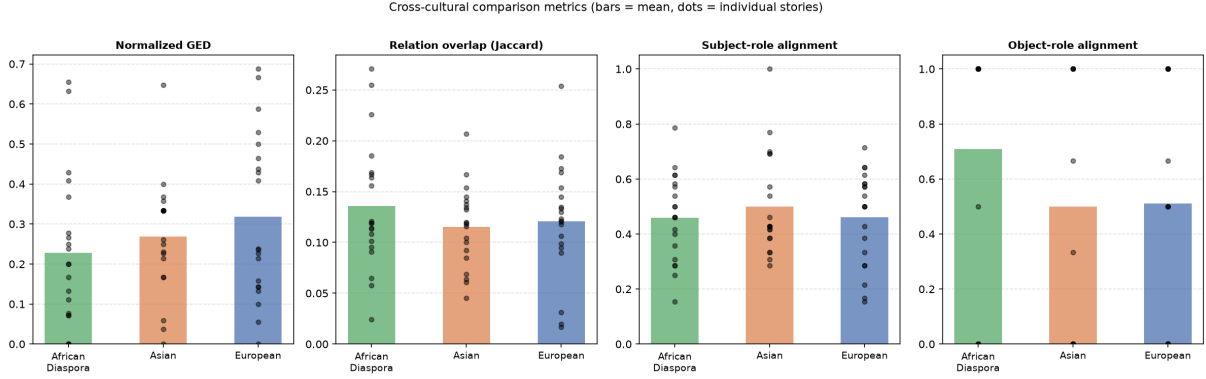


Figure 2: Four structural metrics by target culture. Bars show the mean; dots show individual story pairs.

4.2 Cross-Cultural Structural Comparison

Table 2 summarises the four metrics averaged over all story pairs for each target culture; Figure 2 shows the distributions.

Target culture	Norm. GED	Rel. overlap	Subj. align.	Obj. align.
African Diaspora	0.228	0.136	0.460	0.708
Asian	0.269	0.115	0.500	0.500
European	0.318	0.121	0.461	0.512

Table 2: Cross-cultural comparison metrics (means across 20 story pairs per target culture). Bold marks the best value per column. Lower GED = closer to original; higher overlap/alignment = closer.

Three findings stand out.

European retellings diverge most. The normalised GED for European retellings (0.318) is substantially higher than for African Diaspora (0.228) and Asian (0.269) retellings, indicating that the model restructures the narrative graph most aggressively when targeting a European style.

African Diaspora retellings stay closest. This culture has the lowest GED, the highest relation overlap, and the highest object-role alignment (0.708). The model appears to make smaller structural changes when retelling in an African Diaspora style, possibly because the training distribution for that style is closer to the source stories.

Subject-role alignment is culture-invariant. Across all three target cultures, the fraction of event positions where the protagonist role is preserved hovers between 0.46 and 0.50. The model consistently identifies *who* the story is about, regardless of the target style. Cultural differences surface mainly in secondary relationships (object-role alignment).

Figure 3 shows example narrative graphs for two original stories, illustrating the range of extracted structures: story 9 yields a rich multi-entity graph with many cross-entity edges, while story 1 is dominated by self-loops (events with no clear object entity) and disconnected nodes a known source of noise in LLM-extracted graphs.

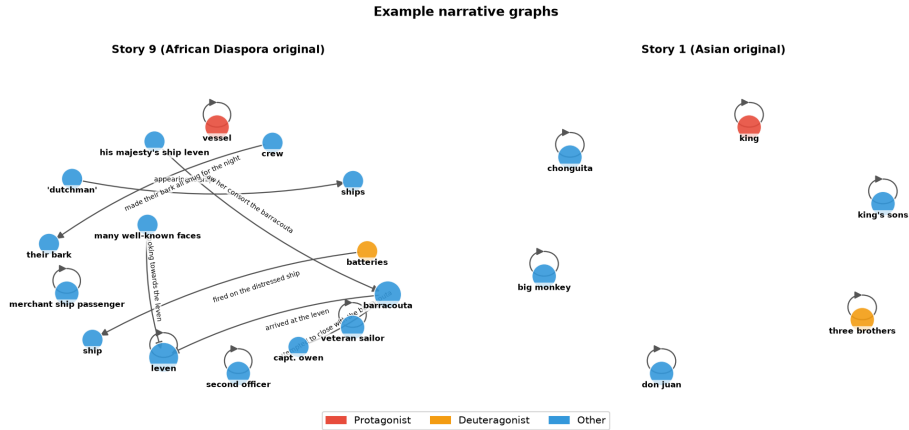


Figure 3: Narrative graphs for two original stories. Node colour indicates role: red = Protagonist, orange = Deuteragonist, blue = Other. Curved loop arcs are self-loops, i.e. events with no clear object entity.

4.3 Semantic Similarity

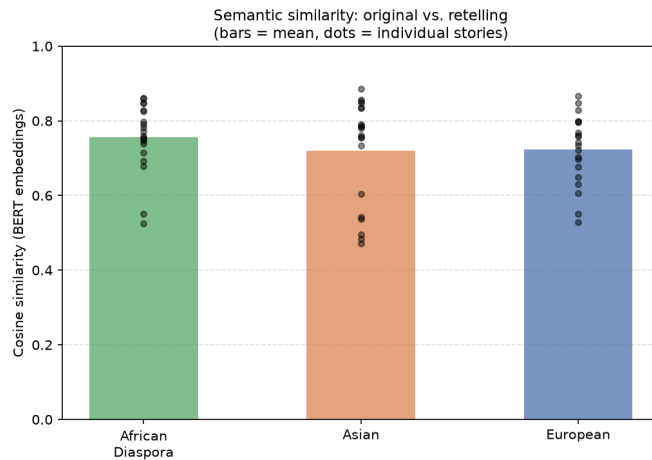


Figure 4: BERT cosine similarity (all-MiniLM-L6-v2) between original and retelling texts. Bars show the mean; dots show individual pairs.

Table 3 and Figure 4 show semantic similarity computed from sentence embeddings.

The BERT results corroborate the graph-structural findings: African Diaspora retellings are the most semantically similar to their originals (0.756), while European and Asian retellings are nearly indistinguishable from each other (≈ 0.72).

A notable observation is the small *spread* of the difference: 0.72 vs. 0.76 is a modest gap, suggesting that on average the model does not drastically alter the meaning of a story regardless of the target culture. Individual variation is substantial, however story 25 scores 0.47 to Asian but 0.79 to African Diaspora, while story 17 achieves 0.89 to Asian. This variance indicates that the model’s cultural adaptation is story-dependent, not uniform.

Target culture	Mean cosine similarity
African Diaspora	0.756
European	0.723
Asian	0.720

Table 3: Mean BERT cosine similarity between original and retelling.

5 Discussion

The clearest result is the asymmetry between target cultures in structural divergence. European-style retellings change narrative graphs most, while African Diaspora-style retellings change them least. One interpretation is that the model has less distinctive “European folktale” prior to draw on and compensates by restructuring the plot; alternatively, the European style may require more assertive changes to vocabulary and setting that indirectly affect the graph. Resolving this would require ablating the prompt or inspecting the model’s attention patterns with explainability techniques.

The clustering finding that original stories do not separate by culture is itself informative. Eight simple graph statistics are insufficient to distinguish European from Asian from African Diaspora folktales. Either the cultural signal lies in subtler structural features (motif sequences, character type distributions) or the structural universality hypothesis has some empirical grounding even in this small corpus.

The stability of subject-role alignment (protagonist identification) across target cultures is reassuring: the model does not confuse *who* the story is about, even when it changes *what* happens. This is consistent with findings in the LLM narrative generation literature that models reliably track primary agents.

Limitations

- **Small corpus.** 30 stories / 60 pairs is too few for statistical testing; reported values are descriptive trends.
- **English only.** All texts are in English, which may bias the model’s cultural adaptations toward the most common English-language folktale conventions.
- **Single model.** Results apply specifically to Llama 3.1 8B and may not generalise to larger or differently-trained models.
- **LLM-extracted graphs.** Events, triples, and entity labels all come from single-pass Llama outputs with no human verification; some noise is expected.
- **GED approximation.** The 10-second timeout means that very large graph pairs may return an approximate rather than exact distance.

6 Conclusions

We presented a pipeline for comparing the narrative structure of folktale originals and LLM retellings across three cultural traditions. Using a combination of narrative graphs and BERT sentence embeddings, we found that Llama 3.1 8B preserves protagonist identity robustly across all target styles, but diverges most in graph structure when targeting European styles and least when targeting African Diaspora styles. Original stories, clustered by graph-level features alone, do not separate by culture, hinting at shared narrative archetypes.

AI Usage Disclaimer

Parts of this project have been developed with the assistance of Antropic's Claude(Opus 4.8). The AI was used to support the development of project ideas, the structuring of methodological workflows, the drafting of descriptive texts, and the identification of relevant datasets and references. All content produced with AI assistance has been carefully reviewed, edited, and validated by me. I take full responsibility for the final content and its accuracy, relevance, and academic integrity.

References

- [1] V. Propp (1928). *Morphology of the Folktale*. Leningrad: Academia.
- [2] C. Brémond (1966). “La logique des possibles narratifs.” *Communications*, 8, 60–76.
- [3] N. Reimers and I. Gurevych (2019). “Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks.” *Proceedings of EMNLP 2019*.
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- [6] A. Hagberg, P. Swart, and D. Schult (2008). “Exploring Network Structure, Dynamics, and Function using NetworkX.” *Proceedings of the 7th Python in Science Conference*.